

“Manski Model”

$$y = \rho Wy + X\beta + WX\theta + u, \quad u = \lambda Wu + \varepsilon$$

$\theta = 0$ Kelejian-Prucha, SARAR, SAC, Cliff-Ord

$$y = \rho Wy + X\beta + u, \quad u = \lambda Wu + \varepsilon$$

$\rho=0$, Spatial Error Model

$$y = X\beta + u, \quad u = \lambda Wu + \varepsilon$$

$\lambda = 0$, Spatial Lag, Lag Y, SAR

$$y = \rho Wy + X\beta + \varepsilon$$

OLS

$\lambda = 0$, Spatial Durbin Model (SDM)

$$y = \rho Wy + X\beta + WX\theta + \varepsilon$$

$\rho = 0$

Spatially Lagged X (SLX)

$$y = X\beta + WX\theta + \varepsilon$$

$\theta = 0$

Spatial Lag, Lag Y, SAR

$$y = \rho Wy + X\beta + \varepsilon$$

$\theta = -\rho\beta$

Spatial Error Model

$$y = X\beta + u, \quad u = \lambda Wu + \varepsilon$$

OLS

$\rho=0$, Spatial Durbin Error Model

$$y = X\beta + WX\theta + u, \quad u = \lambda Wu + \varepsilon$$

$\theta = 0$

Spatial Error Model (SEM)

$$y = X\beta + u, \quad u = \lambda Wu + \varepsilon$$

$\lambda = 0$

Spatially Lagged X (SLX)

$$y = X\beta + WX\theta + \varepsilon$$

OLS

The “Story” behind the models, inspired by Elhorst (2014) and LeSage and Pace (2009), LeSage (2014):

LeSage’s (2014) method: Ask yourself: Should I use a Local Model, or a Global Model?

Local: Neighbors only affect neighbors- no propagation throughout the system (feedback effects). Lag X and Lag errors are local. **Choose SDEM**

Global: A shock to one region affects its neighbors, which affects its neighbors, and so on; this “spatial ripple” also feeds back upon the original region. Any model with a Lag Y is global. **Choose SDM**

Including a lag Y (endogenous interaction effects): Possible space-time dependence (e.g. drug use spreads through space over time), values of Y are jointly determined with that of neighboring agents, or there are “spillover effects”.

Spatial Durbin Model: Can be motivated by assuming that there is a spatially correlated omitted variable (e.g. environmental amenities, highway accessibility, community social capital)

Spatial Error Model: Can be motivated by spatial heterogeneity (e.g., a spatially correlated fixed effect as in panel data models), an omitted variable with spatially distinct effects, others.

Spatially Lagged X: If explanatory variables of area *i* might affect its neighbor *j*

Elhorst, J.P. (2014) *Spatial Econometrics: From Cross-Sectional Data to Spatial Panels*, Springer.

Lesage, James and R. Kelly Pace (2009) *Introduction to Spatial Econometrics*, CRC Press/Taylor & Francis Group.

LeSage, James, (2014), What Regional Scientists Need to Know about Spatial Econometrics, *The Review of Regional Studies*, 44, Issue 1, p. 13-32.